COURSE NAME - CERTIFICATE IN PYTHON PROGRAMMING

Course Code	**: CIP**
Duration	**: 3 months course**
Eligibility	**: 10th Pass**
Papers included	**:**
Theory paper 1	**: Principles of Programming**
Theory paper 2	**: Programming in Python**
Lab 1	**: Python Lab + 1 week GUI based project**
Course Fee	**: Rs.8,000/-+ GST**

Detailed Syllabus:

PAPER I - PRINCIPLES OF PROGRAMMING (50 HRS)

Module – I- Programming Practice and Techniques

Introduction— steps in Programming Process— Understanding Program Specification— Design Program Model— Determine correctness of the program— code the program— Test and debug the program— document the program— Structured Programming- Sequence— Selection— Repetition—Criteria for a good program—Program Tools — Flowcharts — Pseudo codes— Algorithm checking Methods— Dry Run— Independent Inspection— Structured Walk-through—Algorithm Development- Decision— Decision symbol— selection Construct— guidelines for if statements—Algorithm Development- Iteration; While loop construct— for loop construct— repeat until construct— Arrays— What is an array— declaring an array— initialising an array— boundaries of an array— Single dimensional array— two dimensional array—multi dimensional array— Subroutines: modular programming— criteria for decomposing larger modules— recursion.

Module II -Programming Practice using C

Program structure —input and output statements— function definition— delimiters— statement terminator— comment lines —libraries— compiling and running—Variables — constants— identifiers— keywords— data types—Operators —Arithmetic operators- binary operators— unary operators —Arithmetic expressions— assignment operators— multiple assignment and short hand operators .Precedence of operators —type casting—Relational operators and expressions— logical operators and expression— if statement— nested if statement— switch case—Loop Structure- the for loop —while loop— do... while loop. Working of nested loops. Jump statements break and continue—Arrays- Array elements and indices— string/char arrays. Array searching –linear and binary search. Sorting techniques selection and bubble sort— multi dimension arrays and matrix examples— Functions- Function declaration— invoking a function— function body— function prototypes. Formal variables— actual variables—Call by value— call by reference—Global variables —local variables and static variables. Working of recursive functions.

Module –III- Object Oriented Programming Concepts

Structured programming and its drawbacks. Object oriented approach and its advantages. Define classes and objects—Properties and methods of a class— Access Modifiers —public and private sections —Classes and Objects: Implementing Data hiding— Data Abstraction and Encapsulation. Memory allocation for an object— The reference object 'this'. Static class members— Object Arrays—Functions— function overloading. Constructors— parameterised constructor— Destructors— Methods with default arguments— constructor overloading— Inheritance: Introduction— Advantages of inheritance— derived and base classes— types of inheritance— inheritance and access control— constructors in inheritance— super key word. Multiple inheritance —Polymorphism and overriding: Introduction— overriding methods— virtual— super key words. abstract classes— interfaces.

Module –IV - Database Concepts

Database Management system introduction— purpose of database —Advantages of DBMS— Database Architecture—Data Models : Object Based Logical Model —Record based Logical Model—Object-based Logical Model :Entity Relationship Data Models— Types of Relationship : ER Diagram Symbols—Record-based Logical Model: Hierarchical Model— Network model— Relational model—Entity— Strong Entity— Weak entity— subtypes— Sub types— super types. Attribute— Keys:- Primary key— Foreign Key— candidate key— Alternate Ker— Composite Key. Relationship— cardinality of relationship— One-to-One— One-to-many— Many-to Many—Relational algebra :Restrict— Project— Product— Union— Intersect— difference— Join— divide—Normalization: Different forms of Normalization. First Normal Form— second Normal Form —Third Normal form and BCNF

Module –V- Practicing MySQL

Introduction—Mysql features— Data Types—Data Definition Commands : Create database— Drop database and select database. Create table— alter table —add constraints and drop table—Data Manipulation Commands : Insert records — delete records and update records—Querying Data : select command with options —Retrieving specific Attributes— retrieving selected Rows— where and order by clauses—Functions— aggregate functions— group by and having clause—Querying data by using Joins and sub queries— inner join— outer join— self join—Creating views — indexers—Stored procedure

PAPER II – PROGRAMMING IN PYTHON (100 HRS)

Module 1

Python Overview: Introduction to Python, Concepts of OOPs, Features, Applications, Installation on different Operating Systems, Python IDE

Working with Python: Basic Syntax, Process of writing, running of a sample program

Datatype / Variables: Variable and Data Types Declaration, Operators - types of operators, operator precedence, Expressions and Statements

Introduction to Flow Controls: Conditional Statements, Loops, Control Statements, Programming using Python conditional and loops block
Introduction to Functions: Basic Concepts of Functions in Python, Creating & Calling a Function

Module 2
String: Accessing Strings, String operations, String functions and methods, working with Strings
List: Introduction, Accessing list, operations, working with lists, list functions and methods
Sets**:** Introduction, Set operations, Methods, Programming
Dictionary: Introduction, Concept of key-value pair, creating, initialising and accessing the elements in a dictionary, Properties and Functions, Working with dictionaries
Tuple: Introduction, Immutable concept, operations, Tuple functions and methods, working with Tuples

Module 3
Functions: Function basics, Function Arguments, passing parameters (default parameter values, keyword arguments), Programming using functions
Modules: Importing Modules, invoking built in functions, functions from math module, Math module, Random module, Programming using modules
Regular Expressions: Match Functions, Search Functions, Pattern Matching, Programs using Regular Expressions

Module 4
File handling: open and close a file, read, write, and append to a file, standard input, output, and error streams, relative and absolute paths.
Exception Handling**:** Exception, Exception Handling, User Defined Exceptions
Python Object Oriented Programming: Python Classes/Objects, Inheritance, Data Hiding, Overriding, Operator Overloading, Simple programming with OOPs concept
Packages in Python: Packages, Programming with external package

Module 5
Python CGI Programming: CGI Introduction, Architecture, Simple programs
Python Network Programming: Introduction, Server/ Client Methods, Sample Programs
Python Database Programming**:** Introduction to Database, Simple Database Operations, Example Programs
Python GUI Programming: Python Libraries to Create Graphical User Interfaces, Tkinter Programming, Tkinter Widgets, Programming using widgets

Module 6
Difference between Python 2 and Python 3
Introduction to Advanced Features**:** Scientific and technical computing packages (Numpy, Scipy), Visualising packages (Matplotlib), Anaconda, Introduction to NLTK

Lab I – Python Lab + 1 week GUI based project (150 HRS)

Programming Exercise on Basics, Array, Operators.
Programming Exercise using Conditional Statements and Control Flow.

Programming Exercise on Strings.
Programming Exercise on Data Structures - list, tuple, dictionary and set.

Exercise Programs using functions
Exercise Programs using Modules
Practise Exercise on Regular Expressions

Practise programs on File Handling and Exceptions
Exercise programs based on OOPs concept.
Exercise programs using Packages

Practise Exercise on Database Programming
Exercise programs on Network programming
Practise exercise on GUI Programming using Tkinter